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ESCAPE-36 · SUBMISSION NO. 483 · CONCEPTS, METHODS & TOOLS

Sample-Efficient Bayesian Process Optimisation with Structure-Aligned, Prior-Informed Deep Gaussian Processes

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BACKGROUND

Bayesian optimisation

When one data point costs a full experiment, you cannot sweep. **Bayesian Optimisation** keeps a probabilistic surrogate — usually a **Gaussian Process**, which returns a prediction **and** a principled uncertainty — updates it by Bayesian inference after every result, and picks the next experiment where expected progress is highest. The usual prior, a generic smoothness kernel, treats the plant as a **black box**.

THE PROBLEM

Plant-scale evaluations are expensive

Optimising a multi-step process at scale means iterating on the real plant — every evaluation is a full run. Black-box Bayesian optimisation is blind to the flowsheet and spends those runs re-learning what is already known. Meanwhile, cheap **low-fidelity data**, such as lab-scale experiments or coarse simulations, exists for many individual steps, and goes unused.

THE APPROACH

Shape the surrogate like the process

A **Deep Gaussian Process** stacks Gaussian Processes in sequence — each layer's probabilistic outputs feed the next. What if those hidden variables carried physical meaning? The idea is to align the DGP layers with the **process steps**, and inject the inputs or control parameters where they physically act. Furthermore, per-step **low-fidelity models** are pinned as prior means. This together with a fully Bayesian treatment stabilises training in the low-data regime.

METHOD · HOW IT WORKS

A structure-aligned, prior-informed deep Gaussian process

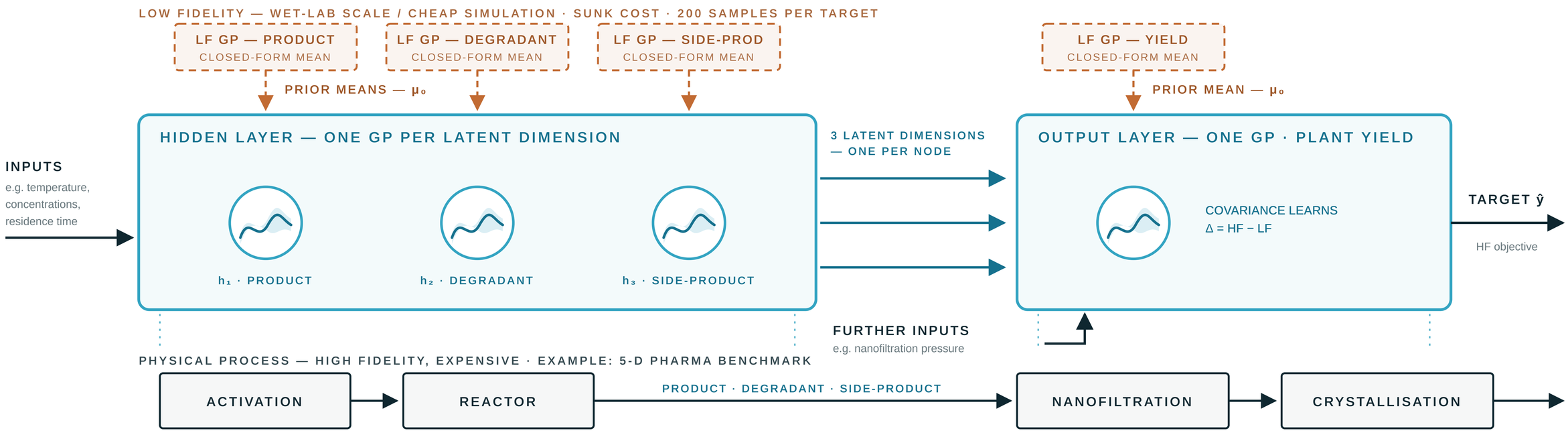


FIG. 01 — STRUCTURE-ALIGNED ARCHITECTURE. We model one or more process steps with one Gaussian Process (GP) layer each (depending on complexity). For the example above (5-D pharma benchmark), the model has two layers representing two process steps each. Low-fidelity (LF) data, such as wet-lab experiments or cheap simulations, is used to pre-train one LF GP per latent dimension; its posterior mean is injected as the prior mean μ_0, and the covariance learns the residual Δ = HF − LF (high − low fidelity). Each layer contains one GP node for each latent dimension (three here). Inputs are injected at the layer where they carry physical meaning.

STRUCTURE ALIGNMENT

One GP per latent dimension

Each hidden unit is an a priori independent GP for one latent dimension; layer width matches the number of intermediate streams. Inputs enter the layer at which they physically act, rather than all at the input layer.

PRIOR INJECTION · M_0

Cheap data as prior mean

For each latent dimension, a low-fidelity GP pre-trained on 200 cheap samples supplies a fixed prior-mean function; the high-fidelity GP then only models the residual Δ between the cheap physics and the real plant.

INFERENCE

Fully Bayesian, via NUTS

Instead of relying on variational inference (VI), the No-U-Turn Sampler (NUTS) is used to draw the hyperparameters and latent function values from the joint posterior, avoiding the variance underestimation typical of VI in the low-data regime.

IMPLEMENTATION

Full-posterior Deep GP in the loop

Sampling runs in **NumPyro/JAX** with 1 000 draws per chain across 32 chains, vectorised in a single MCMC kernel on one GPU. Inferred values cross into **PyTorch**, and a **BoTorch** wrapper enables use within the acquisition function.

SETUP — 10-point initial design · Log-EI, q = 1 · 5 restarts / 64 raw samples · 500 warm-up · Matérn-5/2 in both backends · GPU: H100

RESULTS

Optimisation trajectories on two benchmarks

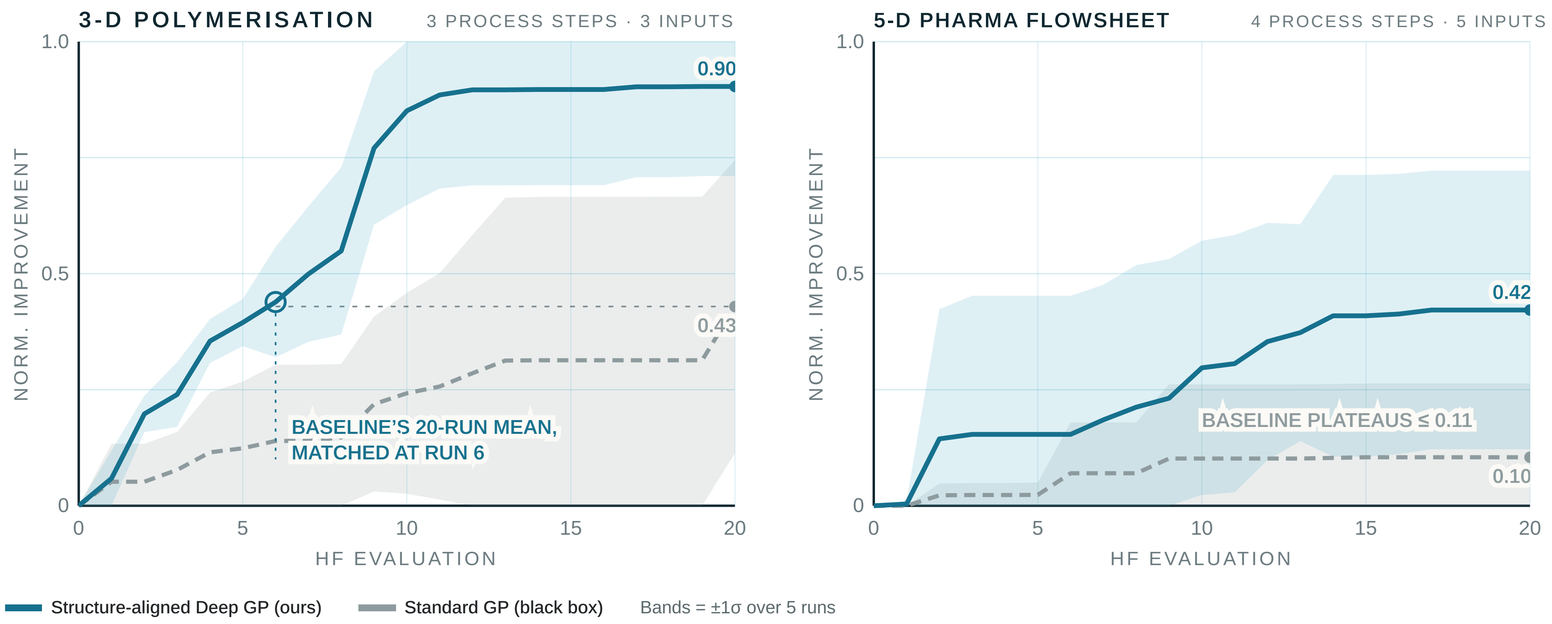


FIG. 02 — OPTIMISATION TRAJECTORIES. Normalised Improvement $(y_i - y_{init})/(y_{best} - y_{init})$, y_{best} = best value found on that benchmark. Sequential selection with Logarithmic Expected Improvement after a 10-point initial design; mean ± 1σ over five runs.

LIMITATIONS

Scope and caveats

- **Compute is the bottleneck.** Each run took roughly eight hours, and training cost scales steeply with the number of data points — capping data set size, depth and model quality.
- **Low-fidelity uncertainty is not propagated.** Its posterior means enter as fixed regularisers; their variance is discarded.
- **Recycle streams are not covered yet.** Both benchmarks are feed-forward; loops would require e.g. additional latent nodes or a fixed-point consistency constraint on the latents.
- **Small evidence base.** Two synthetic benchmark families, n = 5 representative runs each. Comparison with multi-fidelity and grey-box baselines [5, 6, 9] is still open.

POSITIONING

What is new — and what is not

Full Markov-chain Monte Carlo for Deep Gaussian Processes is a deliberate choice, but not a new one [8]. Multi-fidelity Deep Gaussian Processes exist too [9]. The contribution is the **alignment** of process and model topology, as well as the use of low-fidelity prior means to stabilise inference and softly force the latent variables to be physically meaningful.

UP NEXT //

Improving the fully-Bayesian routine would unlock **more data, deeper stacks, and better-resolved models** per iteration.

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